

replace the old fluorescent tube with the new LED tube, keeping the fixture, ballast and all other pre-existing components intact. The tube itself comes with an integrated driver that co-exists with the ballast. This helps to reduce labour costs and efforts, injury risks as well as installation time.

"The installation cost of Type A LED tubes depends on a few factors such as the existing ballast's compatibility with the LED tube," says Rajeev Kr Rohatgi, director, Binay Opto Electronics.

Environment-friendly. LEDs are devoid of mercury or any other harmful chemical, which makes them very safe for use. LED tube lights also substantially reduce the emission of carbon dioxide, ultraviolet rays and other pollutant elements compared to conventional lighting products. Incandescent lights waste 90 per cent of the electricity they consume as heat, producing carbon dioxide in the process. Fluorescent lights, on the other hand, contain mercury—a hazardous liquid metal that can cause some serious health problems if released into the atmosphere.

According to the National UJALA Dashboard website maintained by Energy Efficiency Services Limited and the Ministry of Power, India, as of April 2017, over 1.8 million LED tube lights were distributed throughout India. This initiative resulted in reducing carbon dioxide emission by over 67,000 tonnes and energy consumption by almost 82.4 gigawatt-hours.

Cooling and calming effect. LED tube lights have a high colour rendering index range, which reduces stress on the eyes while working. Additionally, LED tube lights are dimmable, allowing users to set the ambience in their home or office environment. The various colour temperature options determine the intensity of the light output, ranging from reddish, orange, yellow, white up to bluish-white.

Colours emitted by LED tube lights have a very significant effect on the human health and productivity. Bright

bluish-white LEDs have an energising effect on users, while warm yellowish light has a calming effect.

"The colour of white is used for human productivity. The ideal lighting system is one which changes in consonance with the sun's colour. In many offices abroad, when you first enter the office, you find very bright bluish-white lights. This level of brightness at the start of the day acts as your first cup of coffee to kick-start your brain. During lunch hours, the brightness of this light is reduced to warm yellowish colour, to allow the staff to relax. This whole concept of controlling the brightness of light is called 'Human-Centric Circadian Lighting,'" explains Rohatgi.

Challenges

Dependence on ballast. Ironically, the ballast itself becomes a redundant part of the Type A LED tube system. The ballast causes an additional power drainage through the system simultaneously with other components, thus decreasing the tube's efficiency substantially. Plus, the ballast doesn't last as long as the tube, leading to early repair or replacement costs of parts (or even the entire system). Tube replacement is strictly subject to ballast compatibility. Before replacement, users need to ensure that their new tube is compatible with the existing ballast.

Voltage and temperature sensitivity. LEDs require a stable temperature range—a voltage above the threshold value and a current below the indicated rating. Proper heat-sinking setup and electronically controlled voltage passage along with external regulation of voltage and temperature are required for proper functioning of the device. This calls for precision engineering using good-quality driver and fixture components.

While LED tubes seem like a huge capital expenditure, benefits in the long run far surpass the initial investment made. After all, at the end it is all about making your business smarter. **EFY**

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